

Divyat Mahajan

Final Year Ph.D. Candidate
Mila, Université de Montréal

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Google Scholar, Github, LinkedIn

EDUCATION

Université de Montréal/Mila
Ph.D. in Computer Science
Advisor: [Prof. Ioannis Mitliagkas](#)

September '21 - September '26 (Expected)
GPA: 4.3/4.3

Indian Institute of Technology Kanpur
B.S. in Mathematics and Scientific Computing
B.Tech in Computer Science and Engineering (Double Major)

July '14 - June '19
GPA: 8.6/10

RESEARCH INTERESTS

Primary: Pretraining, Post-training, AutoResearch Agents
Others: In-context Learning, Compositional Generalization, Alignment & Safety

WORK EXPERIENCE

• **Meta Superintelligence Labs (FAIR) - Visiting Researcher**
Advisor: [Dr. Kartik Ahuja](#), [Dr. Pascal Vincent](#), and [Dr. David Lopez-Paz](#)

Montréal
January '24 - January '26

→ Developed novel pretraining objectives for LLMs that outperform next-token and multi-token prediction on downstream reasoning tasks

• **Microsoft Research - Research Intern**
Advisor: [Dr. Meyer Scetbon](#) and [Dr. Cheng Zhang](#)

Cambridge
June '23 - September '23

→ Developed transformer-based architectures for in-context learning of causal models, enabling interventional data generation for new tasks at inference time without parameter updates

• **Microsoft Research - Research Fellow**
Advisor: [Dr. Amit Sharma](#)

Bangalore
July '19 - July '21

→ Developed causal inference-inspired techniques to improve out-of-distribution generalization and interpretability of neural networks

SELECTED PUBLICATIONS (*: EQUAL CONTRIBUTION)

• **Iterative Amortized Inference: Unifying In-Context Learning and Learned Optimizers**
[Sarthak Mittal](#), [Divyat Mahajan](#), [Guillaume Lajoie](#), [Mohammad Pezeshki](#)
Under Review

• **Beyond Multi-Token Prediction: Pretraining LLMs with Future Summaries**
[Divyat Mahajan](#), [Sachin Goyal](#), [Badr Youbi Idrissi](#), [Mohammad Pezeshki](#), [Ioannis Mitliagkas](#),
[David Lopez-Paz](#), [Kartik Ahuja](#)
Proceedings of the International Conference of Learning Algorithms (ICLR '26)

• **Compositional Risk Minimization**
[Divyat Mahajan](#), [Mohammad Pezeshki](#), [Charles Arnal](#), [Ioannis Mitliagkas](#), [Kartik Ahuja](#), [Pascal Vincent](#)
Proceedings of the International Conference on Machine Learning (ICML '25)

• **Amortized Inference of Causal Models via Conditional Fixed-Point Iterations**
[Divyat Mahajan*](#), [Jannes Gladrow](#), [Agrin Hilmkil](#), [Cheng Zhang](#), [Meyer Scetbon*](#)
Proceedings of the Transactions on Machine Learning Research (TMLR '25) (J2C Certification), (ICLR '26)

• **Empirical Analysis of Model Selection for Heterogeneous Causal Effect Estimation**
[Divyat Mahajan](#), [Ioannis Mitliagkas](#), [Brady Neal](#), [Vasilis Syrgkanis](#)
Proceedings of the International Conference of Learning Algorithms (ICLR '24) (Spotlight)

• **Additive Decoders for Latent Variables Identification and Cartesian-Product Extrapolation**
[Sébastien Lachapelle*](#), [Divyat Mahajan*](#), [Ioannis Mitliagkas](#), [Simon Lacoste-Julien](#)
Proceedings of the Neural Information Processing Systems (NeurIPS '23) (Oral)

- **Interventional Causal Representation Learning**

Kartik Ahuja, [Divyat Mahajan](#), Yixin Wang, Yoshua Bengio

Proceedings of the International Conference on Machine Learning (ICML '23) (Oral)

- **Domain Generalization using Causal Matching**

[Divyat Mahajan](#), Shruti Tople, Amit Sharma

Proceedings of the International Conference on Machine Learning (ICML '21) (Oral)

- **Split-Treatment Analysis to Rank Heterogeneous Causal Effects for Prospective Interventions**

Yanbo Xu, [Divyat Mahajan](#), Liz Manrao, Amit Sharma, Emre Kiciman

Proceedings of the ACM International Conference on Web Search and Data Mining (WSDM '21) (Oral)

RESEARCH TALKS

- **Compositional Risk Minimization**

Imperial College London, King's College London

May' 25

- **Latent Identification and Extrapolation with Additive Decoders**

Microsoft Research India

January' 24

- **Latent Identification via Interventions**

Microsoft Research Cambridge

July'23

- **Latent Identification in Multi-Task Learning**

Microsoft Research India

Causality Discussion Group, <https://www.matej-zecevic.de/cdg/> [Recording]

August '22

November '23

SOFTWARE

- **RobustDG - Microsoft** [Github] [Commit History]

Core developer for Microsoft's open-source framework for building robust ML models that generalize to unseen domains

- **DiCE - InterpretML** [Github] [Commit History]

Collaborating on the InterpretML's open-source framework to support feasible counterfactual explanation approaches

ACADEMIC SERVICE

- **Reviewer:** ICML 2022-2025, NeurIPS 2021-2025, ICLR 2024-2026, AISTats 2024, TMLR 2025

CleaR 2024-2025, ML RC 2021-2022, MAIS 2022, IEEE SMDS 2021

STUDENT MENTORSHIP

- **Yukti Mukhija** (Ph.D. at Mila, University of Montreal)

Finite Sample Analysis of Compositional Risk Minimization

2025

- **Tejas Kasetty** (M.Sc. at Mila, University of Montreal)

Evaluating Interventional Reasoning Capabilities of Large Language Models

2023-24

AWARDS & ACHIEVEMENTS

- **Outstanding/Top Reviewer Award**

ML RC 2021, ML RC 2022, ICML 2022, NeurIPS 2022, NeurIPS 2024

- **FRQNT Doctorate Scholarship (\$100,000)**

Rank #1 among all applicants in my category. Competition 2024-2025. [DOI]

- **Excellence Scholarship, University of Montreal (\$5,000)**

Awarded for the session 2025-2026 for exceptional academic performance in the Ph.D. program

- **Fast Track Scholarship, University of Montreal (\$14,000)**

Awarded for exceptional academic performance upon transition to Ph.D. from M.Sc. program in Fall 2022

- **Academic Excellence Award, IIT Kanpur**

Awarded for the session 2018-2019 for exceptional academic performance in the undergraduate program